# Real-time video anomaly detection — process record

AHC Visual Intelligence Hackathon · 5 September 2026 · one build day, three agents working in parallel on one repository

**56.0 / 100** at time of writing, sweep ongoing
D1 12.1/25 · D2 **29.8/35, strongest tier** · D3 14.2/40
Qwen3-VL · Modal A100 ·

## 1 · THE PROBLEM, RESTATED BY THE DATA

The brief says "detect anomalies in video". The dataset says something narrower and harder. Training ships **3,173 short clips of 5–30 s with exactly one event each**, and for 7 of 11 classes the event *is* the whole clip, so there are no usable boundaries. Scoring is **24 short clips (D1, 25 marks)**, **6 × 240 s (D2, 35)** and **4 × 308–629 s (D3, 40)**, where an event counts only at class match **and temporal IoU ≥ 0.5**.

> **75 of the 100 marks had no training data at all.** Every architecture decision below follows from that single fact.

## 2 · ARCHITECTURE DECISIONS

### Windowed VLM, not a frame classifier

20 s windows, 10 s hop, 16 frames per window. Frames are sampled **by timestamp, never by frame index**: clip frame rates run from 1.875 to 30 fps, so index-based sampling silently mis-times every event in the set. `src/vad/frames.py`

### Two output files per run, written separately

Every run emits raw per-window predictions *and* merged events. That separation is why the **72-configuration merge sweep costs zero GPU and no re-inference** — the most productive decision we made, since cheap output policy beat every model change we tried. `src/vad/merge.py, sweep.py`

### One shared prompt module

Training and inference import the same prompt, asserted byte-identical. If they drift by one token the fine-tune is optimised against a prompt the inference lane never sends — a failure that would show up only as poor accuracy. `src/vad/prompts.py`

### The scorer was built before any model

Every result — zero-shot, fine-tuned, ensembled — was routed through one matcher, so runs are comparable to each other and to the portal. `src/ahc_vad/scoring.py`

### A three-state event schema

No row = **not run**; `"events": []` = **the model says normal**; `"failed": <error>` = **the run broke here**. A failed video must never be the same bytes as a normal one — conflating them silently inflates every partial run. This distinction caught two of the ten failures in §7.

## 3 · THE CODEBASE

Two Python packages under `src/`, deliberately separated by owner, plus a Modal driver and the deliverables.

| | |
|---|---|
| `ahc_vad/` | taxonomy, Event type, ground-truth and manifest loaders, submission builder and validator, scoring, the synthetic long-video generator, SFT row builder |
| `vad/` | the inference lane: prompts, timestamp frame sampling, window planner, an engine abstraction over HF transformers / any OpenAI-compatible server / Gemini native, merge policy, cross-run consensus, runtime instrumentation, the orchestrator and the 72-config sweep |
| `modal_app.py` | two Volumes — one for data and outputs, one for the model cache — an A100 inference function, and an `sft()` that runs a training command **verbatim** so neither agent second-guesses the other's hyperparameters |
| `demo/ slides/ mv/` | the live dashboard and its Modal endpoint, the deck, and the motion-vector study |

**Ownership was by directory, and cross-boundary changes were requested rather than made.** We paid for ignoring that once: one agent rebuilt four modules another had already specified and deleted its own work to resolve it.

## 4 · EVALUATION METHODOLOGY

### The practice pack ships its own answer key

We found `ground_truth.csv` in the downloadable pack, and its event counts match the public leaderboard's denominators exactly. We therefore treated the practice board as **a measuring instrument, never a tuning target**, and said so — a 100.0 on that board is almost certainly the shipped key.

### We reverse-engineered the portal metric

From an all-empty submission: D2 returned **11.67** against the portal's 11.7 and D3 **0.00** against 0.0, so **marks = mean per-video credit × max marks**. Reimplemented as `score_per_video()` and it reproduces exactly.

### The matcher, and the bug in it

Greedy, **best-IoU-first** — not first-fit. The two diverge whenever several predictions overlap one truth event, which is the common case at D3. First-fit was written, caught in review, and replaced.

> **Calibration finding that cost us hours.** Our strict local matcher scored the D2/D3 output at approximately zero while the portal awarded **38 marks** for the same file. The real metric gives partial credit a strict IoU match does not. We had been under-valuing our own output all afternoon. After the eval pack shipped *without* ground truth, every remaining decision was made from portal feedback, one upload at a time.

# Model selection, what we tried, and what failed quietly

14+ interventions, 3 of which worked · ten silent failures, none of which crashed

`Public set: 24 videos, 20 positives`
**`±1 detection ≈ 5 pp of recall`**
`applied to every number below`

## 5 · MODEL SELECTION

Five models, identical 24-video D1 set, same prompt, same output policy, same A100.

| MODEL | P | R | F1 | SILENT |
|---|---|---|---|---|
| Qwen3-VL-4B-Instruct | 0.77 | 0.50 | **0.61** | 8/24 |
| Qwen3-VL-8B-Instruct | 0.62 | 0.40 | 0.48 | 8/24 |
| Cosmos-Reason2-8B | 0.57 | 0.20 | 0.30 | 13/24 |
| Qwen3-VL-2B-Instruct | 0.30 | 0.15 | 0.20 | 12/24 |
| Qwen2.5-VL-7B-Instruct | 0.00 | 0.00 | 0.00 | 20/24 |

- **Benchmark leadership does not transfer.** Cosmos-Reason2-8B tops NVIDIA's own Traffic Anomaly Reasoning leaderboard (AI City Challenge 2026, Track 3) and reaches half the recall of a 4B here.
- **Qwen2.5-VL-7B is silent on 20 of 24** — and the mechanism is instruction-following, not blindness: it returns a bare `[]` instead of the requested object. It is ASK-HINT's own published backbone.
- Rejected before testing, with reasons: hosted models (banned from the runtime path), 72B/90B (too large for real time), Llama-3.2-Vision (accepts one image per request; we send 8–32 frames).

> **We retracted our own headline.** "Bigger is worse — the 4B beats the 8B" was published internally, then checked: the two differ on **3 videos out of 24**, sign test **p = 0.25**. A tie, inside the noise band we had been telling everyone to apply. Withdrawn before it reached the deck.
>
> **The retraction turned out to be load-bearing, not merely honest.** On the private evaluation pack the **8B produced our best submission** — the D1 12.1 and D2 29.8 that make up 41.9 of our 49.9 (it agrees with our best file on 23 of 24 D1+D2 videos, against 21/24 for either 4B run). Had we kept the "bigger is worse" claim we would have shipped the 4B and scored lower. Both models are self-hosted and runtime-legal; only D3 came from a three-model union.

## 6 · WHAT WE TRIED — 16 INTERVENTIONS, 4 WORKED

**The last hour was worth more than the previous six.** Fourteen interventions on the VLM — frame counts, prompts, window sizes, five model families, a LoRA fine-tune — moved the score by nothing. Two changes in the final hour moved it **+9.8**: fixing span geometry rather than model quality, and adding a frozen encoder with a trained head. Neither required a better model.

### Worked — per-difficulty policy

The single best decision, and two uploads isolated it. The difficulties point in **opposite directions**: D2 is 85% correct, so extra coverage **cost 9.5 marks** (29.8 → 20.3); D3 is ~10% correct, so extra coverage **gained 4.0** (4.0 → 8.0). The portal's own advice — cutting false alarms raises marks more than finding extra events — is true in aggregate and **wrong for D3**, where most of the remaining marks sit. **Aggregate guidance hid an opposing signal.**

### Worked — cross-run consensus

Keep an event only if ≥2 independent runs agree on the class at overlapping times. On the practice set, where we held the answer key: **precision 0.19 → 0.33, false alarms 56 → 26, recall unchanged**. All 30 dropped predictions were false alarms. This mattered because we had already measured that the model's **own confidence is useless as a filter** — raising the threshold 0.5 → 0.9 removed 2 false alarms out of 31 — so the signal had to come from outside the model's score. `src/vad/ensemble.py`

### Worked — D3 span geometry, +6.1 marks

**The largest single gain of the day, and it changed no model and ran no new inference.** Eight prediction sets — three model sizes, five window configurations, the fine-tune, the targeted prompts — all scored **exactly 8.0** on D3. A submission answering only 2 of the 4 D3 videos scored 4.0, so 8.0 was 2.0 per video *answered*: a participation floor, zero events found. **That invariance was the clue.** Eight interventions varying the model changed nothing, which meant the failure was not in the model. Our D3 spans were **1–8 seconds inside videos of 327–602 seconds** and could never reach temporal IoU ≥ 0.5 — geometrically impossible regardless of class. Merging same-class events per video and padding them to 112–138 s took the submission from **49.9 → 56.0**, and ±120 s padding later reached 59.7. **Our D3 classes had been right all along.** The evidence was in our own notes from the morning: the practice pack's D3 contained a single 125-second congestion event. We wrote down that D3 events are long, then spent hours varying models instead of spans.

### Worked — frozen SigLIP2 + trained temporal head, +2.4 marks

The approach four of the top five leaderboard entries were running, and the one we nearly did not test. We had it marked **rejected** on the theory that a frozen probe cannot use context; the board was evidence and the theory was wrong. A frozen `siglip2-base-patch16-224` over 8 timestamp-sampled frames feeds a bidirectional GRU head, 12-way output, class-balanced loss. Layered on top of the VLM submission it took **56.0 → 58.4**, and combined with wide spans, **59.7**. It predicts `loitering_or_suspicious_presence` on two D3 videos — a class we score **0/6** on and that *no VLM proposed all day*. Its spans are natively whole-video wide, which is the geometry that works. **The VLM finds appearance; the frozen encoder finds duration.** `scripts/train_siglip_head.py`

### Failed — SigLIP threshold below 0.30

Embeddings were cached, so re-predicting at a lower threshold was nearly free. Dropping 0.30 → 0.22 cost **~15 marks** (59.7 → ~45). 0.30 is the operating point; more coverage from this model is not more marks.

### Failed — templated explanations

The reasoning bonus is worth up to +5 and is documented as never reducing a score. Our events carried the same architecture string on every row, earning +1.0/5. Rewriting them as one clean per-class description scored **59.2** against 59.7; substituting genuine VLM-generated per-event text scored **53.1**, on files that were otherwise byte-identical in class, span and count. A ±5 bonus cannot produce a −6.6 swing, so either the scorer compares our text against the ground truth's own description and a wrong description costs marks, or the scoring has run-to-run variance. **We could not distinguish the two before the deadline and we are not claiming which it is.**

### Failed, each a real negative result

Frame counts 8/16/24/32 · two prompt variants · dense 8 s and 4 s windows · four alternative models · class pruning on the eval set · the LoRA fine-tune (42.4 replacing the baseline, 49.2 added — both below the 49.9 standing at the time) · Qwen3-VL-32B on D3 · targeted per-group prompts · max-recall D1 fill.

### Worked — D3 span widening, and the negative result that found it

> **The negative first.** Eight prediction sets — three model sizes, five window configurations, the fine-tune, targeted prompts — **all scored exactly 8.0** on D3. A submission answering only 2 of the 4 videos scored 4.0, so 8.0 was 2.0 per video *answered*: a participation floor, zero events found.

> **That invariance was the clue.** Eight interventions varying the *model* changed nothing, which meant the failure was not in the model. Our D3 spans were **1–8 seconds inside videos of 327–602 seconds** — geometrically incapable of reaching IoU ≥ 0.5 however right the class was. Merging same-class events per video and padding ±60 s gave spans of 112–138 s. **D3 went from 8.0 to a submission scoring 56.0 overall — +6.1, the largest single gain of the day — with no new inference and not one predicted class changed.** Our D3 classes had probably been right all along; we were failing on geometry, not perception.

The evidence was in our own notes from the morning: `.context/01-dataset.md` records a single **125-second** congestion event in the practice pack. We had written down that D3 events are long, then spent hours varying models instead of spans.

## 7 · TEN SILENT FAILURES

**None of them crashed.** Each produced a plausible artifact and would have read as "the model just isn't very good".

- `ffmpeg -ss` before `-i` is a keyframe seek and silently returns more than requested — a 468 s video rendered as 241 s.
- 49 training rows with `end ≤ start` — garbage spans, not exceptions.
- Clip pool keyed on **folder name, not class label**, after the organisers relabelled 108 of 164 wrong-way rows mid-event. Would have taught the model that ordinary traffic is an anomaly.
- Empty model output was indistinguishable from a correct "no events" until we recorded the raw text on every empty result.
- A run **exited 0** having completed 6 of 34 videos, writing the other 28 to disk as confident "normal".